



BESIII Analysis Memo

BAM-xxx

BESIII Analysis Memo (January 22, 2026)

Cross section measurement of process $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

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Abstract

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20 **List of changes**

21 For an easy reading of the document, major changes in various versions with respect to the previous
22 one are listed below.

23 **Version 1.0**

- 24 • First version of memo.

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1 Introduction

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2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider—is operating in the τ -charm energy region (2 to 4.9 GeV) with the design luminosity of $1 \times 10^{33} \text{ cm}^{-2} \text{ s}^{-1}$ [1]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric e^+e^- collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the e^+e^- interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [2]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit dE/dx in MDC and flight time in TOF and forms likelihoods $\mathcal{L}(h)(h = K, \pi)$ for a hadron h hypothesis. More details about the BESIII detector are provided in Ref [3].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [4, 5].

$E_{\text{beam}}(\text{GeV})$	Integrated luminosity (pb^{-1})	Data taking time	RunNo
3.773	$2931.8 \pm 0.2 \pm 13.8$ [4]	2010(round03) 2011(round04)	11414 – 13988, 14395 – 14604 20448 – 23454
3.773	4995.0 ± 19.0 [5]	2022(round15)	70522 – 73929
3.773	8157.0 ± 31.0 [5]	2023(round16)	74031 – 78536
3.773	4191.0 ± 16.0 [5]	2024(round17)	78615 – 81094

Table 2: Processes and luminosity scales of the inclusive MC samples used in this analysis. Separate samples are available for each data-taking round. Most processes use samples with an integrated luminosity equivalent to ten times that of the data.

Process	Luminosity scale
$e^+e^- \rightarrow \psi(3770) \rightarrow D^0\bar{D}^0$	$10 \times$
$e^+e^- \rightarrow \psi(3770) \rightarrow D^+D^-$	$10 \times$
$e^+e^- \rightarrow \tau^+\tau^-$	$10 \times$
$e^+e^- \rightarrow \psi(3770) \rightarrow \text{non-}D\bar{D}$	$10 \times$
$e^+e^- \rightarrow q\bar{q}$	$10 \times$
$e^+e^- \rightarrow \gamma_{\text{ISR}}\psi(2S)$	$10 \times$
$e^+e^- \rightarrow \gamma_{\text{ISR}}J/\psi$	$10 \times$
$e^+e^- \rightarrow e^+e^-$	$0.25 \times$
$e^+e^- \rightarrow \gamma\gamma$	$3 \times$
$e^+e^- \rightarrow \mu^+\mu^-$	$15 \times$

3 Data and MC samples

We use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1} , collected by BESIII detector at center-of-mass energy of 3.773 GeV, under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Ref. [4, 5].

Simulated Monte Carlo (MC) samples are employed to study backgrounds, optimize event selection, and estimate detection efficiencies. Two categories of MC samples are utilized in this analysis:

- **The inclusive MC samples** comprehensively model known physics processes at $\sqrt{s} = 3.773 \text{ GeV}$, including $D\bar{D}$ production, $\tau^+\tau^-$ pair production, initial-state-radiation (ISR) production of vector charmonium states (J/ψ and $\psi(2S)$), QED processes, and continuum light hadron production. The charm working group provides separate samples for the different data-taking rounds(round03-04, round15, round16, and round17) [7]. In this analysis, we use inclusive MC samples with an integrated luminosity equivalent to ten times that of the data for most processes, except for the QED processes $e^+e^- \rightarrow e^+e^-$, $\gamma\gamma$, and $\mu^+\mu^-$, where we use the samples with the scaling factors provided by the charm working group. The processes and the corresponding luminosity scales of

the samples used in this analysis are summarized in Table 2.

- **Exclusive signal and background MC samples** generated specifically for this analysis using the PHOKHARA event generator [8, 9, 10] (version 9.1). These samples are produced independently for each data-taking round (round03, round04, round15, round16, round17) to account for variations in detector conditions. Sufficient statistics are generated for each process to ensure reliable efficiency determination and background studies. The generated processes include:

$$- e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\text{ISR}} \text{ (signal process)}$$

$$- e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$$

$$- e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$$

These exclusive samples are used for signal efficiency studies, background characterization, and validation of the analysis methodology.

⁷⁵ **4 Event selections**

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⁷⁷ **5 Summary**

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References

- [1] M. Ablikim *et al.* (BESIII Collaboration), *Future Physics Programme of BESIII*, *Chinese Phys. C* **44** (2020).
- [2] R. X. Yang *et al.*, *MRPC detector for the BESIII E-TOF upgrade*, *Journal of Instrumentation*, **Volume 9**, September 2014. DOI: [10.1088/1748-0221/9/09/C09032](https://doi.org/10.1088/1748-0221/9/09/C09032).
- [3] M. Ablikim *et al.* [BESIII], DOI: [10.1016/j.nima.2009.12.050](https://doi.org/10.1016/j.nima.2009.12.050).
- [4] M. Ablikim *et al.* [BESIII], DOI: [10.1016/j.physletb.2015.11.043](https://doi.org/10.1016/j.physletb.2015.11.043).
- [5] M. Ablikim *et al.* [BESIII Collaboration], DOI: [10.1088/1674-1137/ad70a0](https://doi.org/10.1088/1674-1137/ad70a0).
- [6] S. Agostinelli *et al.* [GEANT4], *Nucl. Instrum. Meth. A* **506**, 250-303 (2003).
- [7] BESIII Collaboration, DocDB: 1483.
- [8] H. Czyż, A. Grzebińska, J. H. Kühn, and G. Rodrigo, DOI: [10.1140/epjc/s2006-02614-7](https://doi.org/10.1140/epjc/s2006-02614-7).
- [9] G. Rodrigo, A. Gehrmann-De Ridder, M. Guillaume, and J. H. Kühn, DOI: [10.1007/s100520100784](https://doi.org/10.1007/s100520100784).
- [10] G. Rodrigo, H. Czyż, J. H. Kühn, and M. Szopa, DOI: [10.1007/s100520200912](https://doi.org/10.1007/s100520200912).

93 **APPENDIX**